

Advanced (Habanero)

- Q1.** What is the slope of the line $y = 2x + 1$?
(A) -1
(B) 1
(C) 2
(D) 0
(E) Undefined
- Q2.** A line has a slope of $\frac{3}{4}$. What is the rate of change?
(A) 3 units per 1 unit
(B) 3 units per 4 units
(C) 4 units per 3 units
(D) 1 unit per 3 units
(E) 4 units per 1 unit
- Q3.** What is the slope of a horizontal line?
(A) -1
(B) 1
(C) 0
(D) Undefined
(E) 2
- Q4.** The table shows the cost of items. What is the rate of change?
(A) \$2 per item
(B) \$1 per item
(C) \$3 per item
(D) \$4 per item
(E) \$5 per item
- Q5.** What is the slope of the line $x = 2$?
(A) -1
(B) 1
(C) 0
(D) 2
(E) Undefined
- Q6.** A line passes through points (1,2) and (3,4). What is the slope?
(A) $\frac{1}{2}$
(B) $\frac{2}{1}$
(C) $\frac{3}{2}$
(D) $\frac{2}{3}$
(E) $\frac{1}{3}$
- Q7.** What is the slope of $y = -x + 1$?
(A) -1
(B) 1
(C) 0
(D) 2
(E) Undefined
- Q8.** A bicycle travels 20 miles in 4 hours. What is the rate of change?
(A) 5 miles per hour
(B) 4 miles per hour
(C) 2 miles per hour
(D) 1 mile per hour
(E) 0 miles per hour

Open Ended Questions (FRQ)

- Q9.** Graph the line $y = -2x + 1$. Find the slope and y -intercept.
- Q10.** A water tank can hold 1000 gallons of water. Water is being filled at a rate of 5 gallons per minute. Write an equation to represent the amount of water in the tank after t minutes. Find the slope and interpret it.

Chef's Tips

- Emphasize the concept of rate of change
- Use real-life examples to illustrate slope
- Have students graph lines to visualize slope